N ~~3~~, ~~4~~, ~~6~~, ~~7~~, ~~8~~, ~~13~~, ~~14~~ Пр 5

N3)

a) $p = 5$ $q = 11$ $e_A = 27$ $M = C = 7$

$N = 55$

$S = C^{d_A} \bmod N_A$ $\varphi(N) = (p-1)(q-1) = 4 \cdot 10 = 40$

$S = 7^{d_A} \bmod 55$

$ed \equiv 1 \bmod \varphi(N)$

$d = e^{-1} \bmod \varphi(N) = 27^{-1} \bmod 40$

$27^{-1} \bmod 40 = 40 \cdot n + 27m = 1$

PAE

$r_0 = 40$ $r_1 = 27$	$r_2 = 40 - 1 \cdot 27 = 13$
$s_0 = 1$ $s_1 = 0$	$s_2 = 1 - 1 \cdot 0 = 1$
$t_0 = 0$ $t_1 = 1$	$t_2 = 0 - 1 \cdot 1 = -1$
$r_3 = 27 - 2 \cdot 13 = 1$	
$s_3 = 0 - 2 \cdot 1 = -2$	$n = s = -2$
$t_3 = 1 - 2(-1) = 3$	$m = t = 3 = d_A$

$S = 7^3 \bmod 55 = 343 \bmod 55 = 13$

b) $p=5$ $q=13$ $e_A=29$ $M=C=10$

$N=65$ $\varphi=4\cdot 12=48$

$29^{-1} \bmod 48 = 48n+29m=1$

$r_0=48$ $r_1=29$	$r_2=48-1\cdot 29=19$
$s_0=1$ $s_1=0$	$s_2=1-1\cdot 0=1$
$t_0=0$ $t_1=1$	$t_2=0-1\cdot 1=-1$
$r_3=29-1\cdot 19=10$	$r_4=19-1\cdot 10=9$
$s_3=0-1\cdot 1=-1$	$s_4=1-1\cdot(-1)=2$
$t_3=1-1\cdot(-1)=2$	$t_4=-1-1\cdot 2=-3$

$r_5=10-1\cdot 9=1$ $\quad n=s=3$

$s_5=-1-1\cdot 2=-3$ $\quad m=t=5=d_A$

$t_5=2-1\cdot(-3)=5$

$10^5 \bmod 65 = 30$

$10^2=100 \bmod 65=35$

$10^4=(35)^2 \bmod 65=1225 \bmod 65=55$

$10^5=55\cdot 10 \bmod 65=30$

b) $p=7$ $q=11$ $N=77$ $e_A=43$

$M=C=5$ $\varphi=60$

$43^{-1} \bmod 60 = 60n+43m=1$

$r_0=60$ $r_1=43$	$r_2=60-1\cdot 43=17$
$s_0=1$ $s_1=0$	$s_2=1-1\cdot 0=1$
$t_0=0$ $t_1=1$	$t_2=0-1\cdot 1=-1$
$r_3=43-2\cdot 17=9$	$r_4=17-1\cdot 9=8$
$s_3=0-2\cdot 1=-2$	$s_4=1-1\cdot(-2)=3$
$t_3=1-2\cdot(-1)=3$	$t_4=-1-1(3)=-4$

$r_5=9-1\cdot 8=1$

$s_5=-2-1\cdot 3=-5$ $\quad n=5$

$t_5=+3-1(-4)=7$ $\quad m=t=d_A=7$

$5^7 \bmod 77$

$5^2=25 \bmod 77=25$

$5^4=(25)^2 \bmod 77=9$

$5^7=5^4\cdot 5^2\cdot 5=9\cdot 25\cdot 5 \bmod 77=47$

$S=47$

№4)

а) $n=55$ $y_A=3$ (7, 28) (22, 15)

$M = S^{y_A} \bmod n$

$28^3 \bmod n$

$28^2 \bmod 55 = 14$

$28^3 \bmod 55 = 14 \cdot 28 \bmod 55 = 7$

верно

$15^2 \bmod 55 = 4$

$15^3 \bmod 55 = 4 \cdot 15 = 5$ неверно

б) $n=65$ $y_A=5$ (5; 42) (10; 30)

$42^5 \bmod n$

$42^2 \bmod 65 = 9$

$42^4 \bmod 65 = 9^2 \bmod 65 = 16$

$42^5 \bmod 65 = 16 \cdot 42 \bmod 65 = 22$

неверно

$30^5 \bmod n$

$30^2 \bmod 65 = 55$

$30^4 \bmod 65 = (55)^2 \bmod 65 = 35$

$30^5 = 35 \cdot 30 \bmod 65 = 10$ верно

в) $n=77$, $y_A=7$ (13, 41) (11; 28)

$41^7 \bmod n$

$41^2 \bmod 77 = 64$

$41^4 \bmod 77 = (64^2) \bmod 77 = 15$

$41^7 = 15 \cdot 64 \cdot 41 \bmod 77 = 13$

верно

$28^7 \bmod n$

$28^2 \bmod 77 = 14$

$28^4 \bmod 77 = (14)^2 \bmod 77 = 42$

$28^7 \bmod 77 = 42 \cdot 14 \cdot 28 \bmod 77 = 63$

неверно

№6) $p=23$ $g=5$

а) $x_A=11$ $r=3$ $M=C=15$

$y_A = g^{x_A} \bmod p = 5^{11} \bmod 23$

$5^2 \bmod 23 = 2$

$5^4 \bmod 23 = 4 \bmod 23 = 4$

$5^8 \bmod 23 = 16$

$5^{11} \bmod 23 = 16 \cdot 2 \cdot 5 \bmod 23 = 22$

$y_A = 22$

$S = r^{-1} u \bmod (p-1)$

$R = g^{r} \bmod p = 5^{3} \bmod 23 =$

$5^{2} \bmod 23 = 2$

$5^{3} \bmod 23 = 2 \cdot 5 = 10$

$R = 10$

$U = (M - x_A \cdot R) \bmod (p-1) = (15 - 11 \cdot 10) \bmod 22 =$

$= -95 \bmod 22 = 15$

$r^{-1} \bmod (p-1) = (p-1) \cdot n + r \cdot m$

$r_0 = 22 \quad r_1 = 3 \quad r_2 = 22 - 3 \cdot 7 = 21$

$s_0 = 1 \quad s_1 = 0 \quad s_2 = 1 - 7 \cdot 0 = 1$

$t_0 = 0 \quad t_1 = 1 \quad t_2 = 0 - 7(-1) = -7$

$S = r^{-1} U \bmod (p-1) = -7 \cdot 15 \bmod 22 =$

$= 5$

Ответ: $y_A = 22 \quad S = 5$

d) $x_A = 10 \quad r = 17 \quad M = C = 5$

$y_A = 5^{10} \bmod 23$

$5^{2} \bmod 23 = 2$

$5^{10} = 5^{8} \cdot 5^{2} \bmod 23 = 16 \cdot 2 \bmod 23 = 9$

$R = 5^{17} \bmod 23$

$5^{17} = 5^{10} \cdot 5^{4} \cdot 5 = 9 \cdot 4 \cdot 5 = 19 \bmod 23$

$U = (M - x_A \cdot R) \bmod (p-1) = 15 - 10 \cdot 19 \bmod 22 =$

$= -185 \bmod 22 = 13$

$r^{-1} \bmod (p-1) = (p-1) n + r \cdot m$

$r_0 = 22 \quad r_1 = 17 \quad r_2 = 22 - 17 = 7$

$s_0 = 1 \quad s_1 = 0 \quad s_2 = 1 - 1 \cdot 0 = 1$

$t_0 = 0 \quad t_1 = 1 \quad t_2 = 0 - 1 \cdot 1 = -1$

$r_3 = 17 - 2 \cdot 7 = 1$

$s_3 = 0 - 2 \cdot 1 = -2$

$t_3 = 1 - 2(-1) = 3$

$S = 3 \cdot 13 \bmod 22 = 17$

$y_A = 9$

b) $x_A = 3 \quad r = 13 \quad M = C = 8$

$y_A = 5^{3} \bmod 23$

$5^{3} \bmod 23 = 10$

$y_A = 10$

$k = 5^{13} \bmod 23 = 9 \cdot 10 \bmod 23 = 21$

$u = (M - x_A \cdot k) \bmod (p-1) = 8 - 3 \cdot 21 \bmod 22 =$

$= -55 \bmod 22 = 11$

$r^{-1} \bmod (p-1) = (p-1) \cdot n + r \cdot m$

$r_0 = 22 \quad r_1 = 13 \quad r_2 = 22 - 1 \cdot 13 = 9$

$s_0 = 1 \quad s_1 = 0 \quad s_2 = 1 - 1 \cdot 0 = 1$

$t_0 = 0 \quad t_1 = 1 \quad t_2 = 0 - 1 \cdot 1 = -1$

$r_3 = 13 - 1 \cdot 9 = 4 \quad r_4 = 9 - 2 \cdot 4 = 1$

$s_3 = 0 - 1 \cdot 1 = -1 \quad s_4 = 1 - 2(-1) = 3$

$t_3 = 1 - 1 \cdot (-1) = 2 \quad t_4 = -1 - 2(2) = -5$

$s = r^{-1} u \bmod (p-1) = -5 \cdot 11 \cdot \bmod 22 =$

$= 11$

$y_A = 10$

№7) $p = 23 \quad g = 5$

а) $y_A = 22 \quad (15; 20; 3) \quad (15; 10; 5)$

$\qquad\qquad\qquad\qquad M \quad R \quad S$

$y_A^R \cdot R^S \equiv g^M \bmod p$

$22^{20} \cdot 20^3 \bmod 23$

$22^2 \bmod 23 = 1$

$22^4 \bmod 23 = 1^2 \bmod 23 = 1$

$22^8 \bmod 23 = 1^2 \bmod 23 = 1$

$22^{20} \bmod 23 = 1$

$20^2 \bmod 23 = 9$

$20^3 \bmod 23 = 9 \cdot 20 \bmod 23 = 19$ ①

$5^{15} \bmod 23$

$5^2 \bmod 23 = 2$

$5^4 \bmod 23 = 4$

$5^8 \bmod 23 = 16$

$5^{15} = 5^8 \cdot 5^4 \cdot 5^2 \cdot 5 = 2 \cdot 4 \cdot 16 \cdot 5 \bmod 23 =$

$= 19$ ②

$19 = 19$ верно

$2^{10} \cdot 10^5 \equiv 5^{15} \bmod 23$

$5^{15} \bmod 23 = 19$

$22^{10} \bmod 23 = 1$

$10^2 \bmod 23 = 8$

$10^4 \bmod 23 = 8^2 \bmod 23 = 18$

$10^5 \bmod 23 = 18 \cdot 10 = 19$

$19 = 19$ верно

d) $y_A = 9$ $(5, 19, 17)$ $(7, 17, 8)$
H R S

$9^{19} \cdot 19^{17} \equiv 5^5 \bmod 23$

$5^5 = 4 \cdot 5 \bmod 23 = 20$

$9^2 \bmod 23 = 12$

$9^4 \bmod 23 = 12^2 \bmod 23 = 6$

$9^8 \bmod 23 = 6^2 \bmod 23 = 13$

$9^{16} \bmod 23 = 13^2 \bmod 23 = 8$

$9^{19} = 8 \cdot 12 \cdot 9 \bmod 23 = 13$ (1)

$19^2 \bmod 23 = 16$

$19^4 \bmod 23 = 16^2 \bmod 23 = 3$

$19^8 \bmod 23 = 3^2 \bmod 23 = 9$

$19^{16} \bmod 23 = 9^2 \bmod 23 = 12$

$19^{17} \bmod 23 = 12 \cdot 19 \bmod 23 = 21$

~~неверно~~ $21 \cdot 13 \bmod 23 = 20$

~~неверно~~

$20 = 20$ верно

$9^{17} \cdot 17^8 \equiv 5^7 \bmod 23$

$5^7 \bmod 23 = 17$

$9^{17} \bmod 23 = 8 \cdot 9 \bmod 23 = 3$

$17^8 \bmod 23$

$17^2 \bmod 23 = 13$

$17^4 \bmod 23 = 13^2 \bmod 23 = 8$

$17^8 \bmod 23 = 8^2 \bmod 23 = 18$

$18 \cdot 3 \bmod 23 = 8$

$8 \neq 17$ неверно

в) $y_A = 10\ (3;\ 17;\ 12)\ (2;\ 17;\ 12)$

$10^{17} \cdot 17^{12} \equiv 5^3 \bmod 23$

$5^3 \bmod 23 = \underline{10}$

$10^{17} \bmod 23$

$10^8 \bmod 23 = 11^2 \bmod 23 = 2$

$10^{16} \bmod 23 = 4$

$10^{17} \bmod 23 = 17$

$17^{12} \bmod 23 = 17 \cdot 7 \bmod 23 = 6$

$17 \cdot 6 \bmod 23 = \underline{10}$

$10 = 10$ верно.

$10^{17} \cdot 17^{12} \equiv 5^2 \bmod 23$

$10 \neq 2$ неверно

№8) $p = 467$ $g = 2$ $x_A = 127$ $M = 100$

$r = 213$

Найти:

1) $y_A = g^{x_A} = ?$

2) подпись сообщ M

3) проверить подпись

3) $y_A = g^{x_A} \bmod p = 2^{127} \bmod 467 =$

$=$

$2^2 \bmod 467 = 2$ $2^8 \bmod 467 = 256$

$2^4 \bmod 467 = 4$ $2^{16} \bmod 467 = 156$

$2^{32} \bmod 467 = 52$ $2^{64} \bmod 467 = 369$

$2^{127} = 369 \cdot 52 \cdot 156 \cdot 256 \cdot 4 \cdot 2 \cdot 2 \bmod 2$

$= \underline{132}$

г) $S = r^{-1} u \bmod (p-1)$

$R = g^r \bmod p$

$u = (M - X_A \cdot R) \bmod (p-1)$

$r^{-1} \bmod (p-1) = (p-1) \cdot n = r \cdot m$

$K = 2^{213} \bmod 467$

$2^{128} \bmod 467 = 264$

$2^{213} \bmod 467 = 264 \cdot 369 \cdot 156 \cdot 42 \bmod 467 =$

$= \underline{29}$

$U = (M - x_A \cdot R) = (100 - 127 \cdot 29) \bmod 466 =$
$\bmod (p-1)$

$= \underline{145}$

$r^{-1} \bmod (p-1) = (p-1) \cdot n + r \cdot m$

$r_0 = 466$ $r_1 = 213$	$r_2 = 466 - 2 \cdot 213 = 40$
$s_0 = 1$ $s_1 = 0$	$s_2 = 1 - 2 \cdot 0 = 1$
$t_0 = 0$ $t_1 = 1$	$t_2 = 0 - 2 \cdot 1 = -2$
$r_3 = 213 - 5 \cdot 40 = 13$	$r_4 = 40 - 3 \cdot 13 = 1$
$s_3 = 0 - 5 \cdot 1 = -5$	$s_4 = 1 - 3(-5) = 16$
$t_3 = 1 - 5 \cdot (-2) = 11$	$t_4 = -2 - 3(11) = -35$

$r^{-1} = -35 \bmod 466 = 431$

$S = 431 \cdot 145 \bmod 466 = 51$

проверка на корректн.

$НОД(p-1; r) = 1$

$НОД(466; 213) = 1 \Rightarrow$ корректн.

3) $y_A = 132$ $(100; 29; 51)$ $g = 2$ $p = 467$

$132^{29} \cdot 29^{51} = 2^{100} \bmod 467$

$\cancel{2^{100} \bmod 467 = 369 \cdot 52 \cdot 4 \bmod 467 =}$

$\cancel{= 270}$

$2^{100} \bmod 467 = 369 \cdot 52 \cdot 16 \bmod 467 =$

$= \underline{189}$

$132^{29} \bmod 467 =$

$132^{2} \bmod 467 = 145$

$132^{4} \bmod 467 = 145^{2} \bmod 467 = 10$

$132^{8} \bmod 467 = 100$

$132^{16} \bmod 467 = 195$

$132^{29} \bmod 467 = 195 \cdot 100 \cdot 10 \cdot 132 \bmod 467 =$
$= \underline{216}$

$29^{51} \bmod 467$

$29^2 \bmod 467 = 374$ $\quad 29^8 \bmod 467 = 207$

$29^4 \bmod 467 = 143$ $\quad 29^{16} \bmod 467 = 352$

$29^{32} \bmod 467 = 149$

$29^{51} \bmod 467 = 149 \cdot 352 \cdot 374 \cdot 29 \bmod 467 =$

$= 176$

$176 \cdot 216 \bmod 467 = 189$

$189 = 189$ верно

№ 13) $p = 67$ $q = 11$ $g = 25$

$y_A = ?$ $\quad s = ?$ $\quad$ а) $x_A = 3$ $M = f(M) = 10$

$k_A = 1$

$y_A = (g^{x_A} \bmod p) \bmod q =$

$= (25^3 \bmod 67) \bmod 11 = 3$

$25^3 \bmod 67 = 14$

$14 \bmod 11 = 3$

$r_A = (g^{k_A} \bmod p) \bmod q =$

$= (25^1 \bmod 67) \bmod 11 = 3$

$s = (k_A^{-1} \cdot F(M) + x_A \cdot r_A) \bmod q =$

$= (1 \cdot 10 + 3 \cdot 3) \bmod 11 = 1$

б) $x_A = 8$ $M = F(M) = 1$ $k_A = 3$

$y_A = (25^8 \bmod 67) \bmod 11 =$

$= 24 \bmod 11 = 2$

$25^6 \bmod 67 = 62$ $\quad 25^8 \bmod 67 = 62 \cdot 25^2 \bmod 67 =$

$= 24$

$r_A = (25^3 \bmod 67) \bmod 11 = 3$

$s = (3 \cdot 1 + 3 \cdot 8) \bmod 11 = 5$

в) $x_A = 5$ $M = F(M) = 5$ $k_A = 9$

$y_A = (25^5 \bmod 67) \bmod 11 = 7$

$25^5 = 14 \cdot 25^2 \bmod 67 = 40$

$r_A = (25^9 \bmod 67) \bmod 11 = 9$

$25^9 \bmod 67 = 40 \cdot 14 \cdot 25 \bmod 67 = 64$

$s = (9 \cdot 5 + 9 \cdot 5) \bmod 11 = 2$

в 14) $p = 67$ $q = 11$ $y = 25$

$y_A = 14$ (10; 4; 5) (10; 7; 5)

M r s

$u_1 = s \cdot (H(M))^{-1} \bmod q$

$u_2 = -r_A \cdot (H(M))^{-1} \bmod q$

$u_1 = 5 \cdot (10)^{-1} \bmod 11 = 6$

$10^{-1} \bmod 11 = 11 \cdot m + 10 \cdot n$

$r_0 = 11$ $r_1 = 10$	$r_2 = 11 - 1 \cdot 10 = 1$
$s_0 = 1$ $s_1 = 0$	$s_2 = 1 - 1 \cdot 0 = 1$
$t_0 = 0$ $t_1 = 1$	$t_2 = 0 - 1 \cdot 1 = -1$

$m = -1 \bmod 11 = 10$

$u_2 = -4 \cdot (10)^{-1} \bmod 11 = 4$

$v = (g^{u_1} \cdot (y_A)^{u_2} \bmod p) \bmod q$

$v = (25^6 \cdot 14^4 \bmod 67) \bmod 11 =$

$= (25 \cdot 62 \bmod 67) \bmod 11 = 7$

$25^6 \bmod 67 = 40 \cdot 25 \bmod 67 = 62$

$14^2 \bmod 67 = 62$ $14^4 = 62^2 \bmod 67 = 25$

$v \neq r$ ($7 \neq 4$ неверно)

$u_1 = 5 \cdot 10^{-1} \bmod 11 = 6$

$u_2 = -7 \cdot (10)^{-1} \bmod 11 = 4$

$v = (25^6 \cdot 14^7 \bmod 67) \bmod 11 =$

$= (62 \cdot 59 \bmod 67) \bmod 11 = 7$

$14^7 \bmod 67 = 25 \cdot 62 \cdot 14 \bmod 67 = 59$

$v = r$ $7 = 7$ верно

б) $y_A = 24$ (1; 3; 5) (1; 4; 3)

$u_1 = 5 \cdot (1)^{-1} \bmod 11 = 5$

$u_2 = -3 \cdot 1 \bmod 11 = 8$

$v = (25^5 \cdot 24^8 \bmod 67) \bmod 11 =$

$= (40 \cdot 64 \bmod 67) \bmod 11 = 3$

$24^2 \bmod 67 = 40$ $24^4 \bmod 67 = 59$

$24^8 \bmod 67 = 59^2 \bmod 67 = 64$

$v = r$ $3 = 3$ верно

$u_1 = 3$

$u_2 = 7$

$V = (25^3 \cdot 24^7 \bmod 67) \bmod 11 = (14 \cdot 25 \bmod 67) \bmod 11 =$

$= \underline{4}$

$24^7 \bmod 67 = 5 \cdot 9 \cdot 40 \cdot 24 \bmod 67 =$

$= 25$

$V = r \quad 4 = 4$ верно

б) $y_A = 40 \quad (10; 7; 4) \quad (7; 9; 2)$

$u_1 = 4 \cdot (10)^{-1} \bmod 11 = \underline{7}$

$u_2 = -7 \cdot (10)^{-1} \bmod 11 = \underline{7}$

$V = (25^7 \cdot (40)^7 \bmod 67) \bmod 11 =$

$= (9 \cdot 22 \bmod 67) \bmod 11 = \underline{9}$

$25^7 \bmod 67 = 62 \cdot 25 \bmod 67 = 9$

$40^2 \bmod 67 = 59 \quad 40^4 \bmod 67 = 64$

$40^7 \bmod 67 = 59 \cdot 64 \cdot 7 \bmod 67 = 22$

$V \neq r \quad 9 \neq 7$ неверн.

$u_1 = 2 \cdot (7)^{-1} \bmod 11 = \underline{5}$

$r_0 = 11 \quad r_1 = 7 \quad | \quad r_2 = 11 - 1 \cdot 7 = 4$

$s_0 = 1 \quad s_1 = 0 \quad | \quad s_2 = 1 - 1 \cdot 0 = 1$

$t_0 = 0 \quad t_1 = 1 \quad | \quad t_2 = 0 - 1 \cdot 1 = -1$

$r_3 = 7 - 1 \cdot 4 = 3 \quad | \quad r_4 = 4 - 1 \cdot 3 = 1$

$s_3 = 0 - 1 \cdot 1 = -1 \quad | \quad s_4 = 1 - 1(-1) = 2$

$t_3 = 1 - 1 \cdot (-1) = 2 \quad | \quad t_4 = -1 - 1(2) = -3$

$m = -3 \bmod 11 = \underline{8}$

$u_2 = -9 (7)^{-1} \bmod 11 = \underline{5}$

$V = (25^5 \cdot 40^5 \bmod 67) \bmod 11 = \underline{2}$

$25^5 \bmod 67 = 40$

$40^5 \bmod 67 = 64 \cdot 40 \bmod 67 = 14$

$(40 \cdot 14) \bmod 67 = 24$

$V \neq r \quad 9 \neq 2$ неверн